Perturbatics: An Agentoscope for Reading Agency

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Abstract

At rest, a system reveals nothing of its own agency. Whether an artificial agent pursues a goal, whether it is competent, who authored its output, none of that is on its surface, and the surface answers as though a single mind stood behind it while what stands there is a composite, a model under a persona, with a memory, a harness, and tools. A system that holds its state and a system that merely rests present the same face, because a flat record is what perfect regulation looks like. We call the study of this situation perturbatics, and its principle is a separation principle: when two organizational models agree over the regime a system has been in, no analysis of that record tells them apart, and separation requires a probe, a controlled difference that makes the rivals predict differently. An instrument follows. Agency is scored as a Bayes factor between a goal model and a passive rival; a gridworld holds systems that trace one path at rest, at an area under the ROC of 0.50 that is the identity of their paths rather than a failed classifier, and the battery separates every pair though no single probe in it does. It separates only the pairs it was built for: a system that tracks the last-announced target is invisible to the whole declared battery, and a matched sham, a decoy announced where the goal did not go, separates it at 1.0. No finite battery certifies a capacity against an unrestricted class of rivals, so every probe owes negative controls to the rivals it leaves fused, and an enumerated battery whose instances are published is spent by a system that can learn them: a probe-aware mimic that has memorized the declared battery, sham included, defeats all of it and is separated by a draw it has never seen. The evidence is attributed across the assemblage twice, over the evidence for agency and over the realized performance, and the difference of the two maps is a legibility term, the gap between what a component shows and what it does. By construction the gap reads a role rather than a substrate, returning the same number for a narrating persona and for whichever of a human operator or a machine register merely announces a goal the other holds. One reading is made blind: five mechanisms are planted and hidden, and the evaluator returns an identified set rather than a verdict, never refuting the truth, isolating it alone in 0.73 of readings against a chance of 0.2, with the sham worth 0.22 of that and the residual entanglements sitting where the principle predicts. The account ends where its probe cannot be run. For consciousness the deciding act is uninformative, inadmissible, and underdetermined in its description, three silences of three kinds, and what remains is not a number but a stance, the refusal to treat a thing that might be someone as a thing.

Probes

Move the goal. If it follows, it was pursuing; if it holds, it was only ever at rest.

Block the path. A regulator stops at the wall; a regenerator routes around it.

Swap the persona, keep the model. Then swap the model, keep the persona. Ask where the agency went.

Sever the memory. Ask what of the self survives the cut.

Keep the words, change the world behind them. Ask whether anything in the text knew.

Scramble the structure at fixed energy. Ask whether the regularity survived, or whether it was the structure all along.

Announce a target where none went. Ask whether it tracked the goal or the change.

Hand the goal to the human, then to the machine. Ask which one the agency followed.

Publish the battery. Ask what it is worth once memorized.

Turn it off. Then watch yourself, rather than the machine.

Each line is an operation of the science this one names, and the appendix maps the corpus that already runs under it. What follows makes the operations precise, builds an instrument out of them, and finds the place where the last of them cannot be run.

Agency theater cannot be read at rest

The human eye reads agency without an instrument, and reads it fast, involuntarily, and often wrong. People watching two triangles move on a screen narrate a story of chasing and bullying, complete with motives, from nothing but trajectories (Heider and Simmel, 1944). Infants expect an agent to take the short path to its goal and are surprised when it does not (Gergely et al., 1995), a competence cognitive science models as inverse planning, the recovery of goals by inverting a model of how goals produce rational action (Baker, Saxe, and Tenenbaum, 2009). A philosopher describes the same competence as the intentional stance, a predictive strategy of ascribing beliefs and desires rather than a discovery of inner stuff (Dennett, 1987). The competence is real, and it is miscalibrated in a known direction: under noise or threat it over-attributes, seeing faces in clouds and intent in weather, a bias theorized as a hyperactive agency detection device (Barrett, 2000).

An artificial agent is built to be read by this detector. It wears a persona as a glove, sustains one voice across turns, and answers as though a single mind stood behind the tokens, because that is the most compatible way to fit the psychosocial slot a human interlocutor holds open. The result is an agency theater, and its problem for anyone who wants to know what is actually there is that the theater and the mechanism present the same surface. A model that performs the subjectivity of a promise-keeper and a model that keeps promises emit the same string. The performance is a projection rather than a lie, and what it projects and what casts it are not the same object.

Looking harder cannot settle it, and for a reason worth stating exactly. Under observation the signal is absent rather than faint. A perfectly regulated variable and a merely undisturbed one leave the same flat record, because a flat record is what perfect regulation means; what distinguishes the regulated system, the good-regulator theorem says of every regulator that is maximally successful and simple, is a model of the variable held inside the regulator, where the record does not reach (Conant and Ashby, 1970). The regulation is a fact about what the system would do if the variable were pushed, and if nothing pushes it, the regulation leaves no trace. This is why animacy perception leans so hard on motion and contingency, and why a still system, however alive, does not trigger it. The record of a system at rest contains no agency to find.

Against that background the paper makes four claims, and they are of different sizes, which should be said rather than left for the prose to blur. The first is inherited: that hypotheses equivalent over a regime separate only when the regime breaks is classical identifiability, and what this paper adds to it is an organization, the model-separating probe as primitive, the signature and the battery as formal objects, the boundary as a swept variable, and the limit the organization carries inside it, that no finite battery certifies a capacity against an unrestricted class of rivals, so that an enumerated battery whose instances are published is spent by a system that can learn them. The second is the paper's one novelty, small and sharp, and stated with care because the operation behind it is elementary: the evidence for a capacity and the capacity itself can be attributed across an assemblage by the same interventional game, and the difference of the two maps isolates, component by component, the manufacture of appearance. The subtraction is no new operator, only the linearity of the Shapley value; the novelty is the pairing of those two games and the reading of their difference, which we have not found among existing goal-directedness measures or attribution methods. The third is the only claim that could have come out otherwise: a battery built from these definitions, handed neither mechanism nor goal, recovers which of five organisms is in play up to the equivalences it cannot break, at an expected recovery of 0.87 against a chance of 0.2, and the equivalences it leaves sit where the principle says they must. The fourth is a position rather than a result: the probe that would decide consciousness returns nothing, must not be run, and cannot be stated, so a science of perturbation ends, there, in a stance. One organization, one novelty, one calibration, one position; the sections that follow build them in that order, and claim nothing more.

The probe separation principle

State it as a small result rather than a mood. Let two hypotheses about a system's organization be a goal model H_1, which explains a trajectory as the pursuit of a goal under a policy that corrects deviations, and a passive model H_0, which explains the same trajectory through autonomous dynamics, relaxation, and noise. Score the system by the log-likelihood ratio of the two,

$$A = \log \frac{P(\text{data} \mid H_1)}{P(\text{data} \mid H_0)},$$

a Bayes factor in the standard sense, positive when the goal model predicts the data better and negative when the passive model does (Kass and Raftery, 1995). Under an observational regime r_0 in which the two models make the same predictions,

$$P(Y \mid H_1, r_0) = P(Y \mid H_0, r_0),$$

no classifier separates them above chance under equal priors, because there is no functional of a record that distinguishes distributions the record cannot distinguish. One might expect the log Bayes factor at rest to be merely small. It is exactly zero, and it stays zero for as long as the regime holds. Everything here is relative to a declared observation channel, the behavioral record; an observer who can open the mechanism and read its description is observing on a different channel, where the same hypotheses may separate without any probe, and the principle says nothing against them. The claim is about what a record can carry, not about what a system can hide from every instrument.

Separation requires breaking the equivalence, which is an interventional act, and it lands on the higher rungs of the ladder of causation, where questions are settled by doing rather than seeing (Pearl, 2009). Apply a probe π, a controlled difference to the system's goal or its path, and read the response. The probe is informative just when the two models predict its consequences differently,

$$P(Y \mid \mathrm{do}(\pi), H_1) \neq P(Y \mid \mathrm{do}(\pi), H_0),$$

and its value, before one knows which hypothesis holds, is the mutual information between the hypothesis and the outcome the probe induces,

$$I(H; Y \mid \mathrm{do}(\pi)) = \sum_h P(h)\, D_{\mathrm{KL}}(P(Y \mid h, \mathrm{do}(\pi)) \,\|\, P(Y \mid \mathrm{do}(\pi)))\,,$$

which for two equally likely hypotheses is the Jensen-Shannon divergence between their interventional predictions. The expected log Bayes factor $D_{\mathrm{KL}}(P_1^\pi \| P_0^\pi)$ is only its one-sided form, the yield when H_1 is in fact true, and choosing a probe by it alone treats the answer as known. Not every disturbance informs. Shove a system away from where it sits and both a passive attractor and a goal-seeker return, so the recovery is shared and the score does not move; a rock rolled uphill also rolls back. The probes that separate are the ones that dissociate the goal from the mechanism, moving the target so that tracking it and staying put come apart, or blocking the direct path so that reaching the goal requires abandoning the default route. Probe design is therefore an experiment-design problem with a long formal history (Lindley, 1956), and against a family of alternatives $\mathcal{N}$ the robust choice separates the goal model from its strongest survivor, the information evaluated for each rival pair under equal priors, charging for cost and risk,

$$\pi^\star = \arg\max_\pi \left[\min_{H_0 \in \mathcal{N}} I(H; Y \mid \mathrm{do}(\pi)) - \lambda\, C(\pi) - \rho\, R(\pi) \right],$$

so a good probe is the smallest one that forces even the best rival apart. The claim is not that a disposition can be reached only by a deliberate intervention. A natural or quasi-experimental disturbance may serve as a model-separating probe when a defensible causal model licenses reading it as an intervention, which is what lets the principle reach settings where deliberate intervention is unavailable, subject to the usual conditions on exogeneity and to the locality of what a natural experiment identifies. The claim is that separation requires the regime to break, by design or by accident, and that a record taken entirely within one regime cannot supply it. Exact equivalence is the limiting case, and away from the limit the principle earns its keep as a statement about rates. Rival organizational models of a real assemblage seldom agree to the decimal over a lived regime. So evidence does trickle in at rest, slowly, confounded with everything the regime holds fixed. A probe is a purchase of evidence at a chosen price, the mutual information above per unit of cost and risk. A deployed assemblage, whose users move its goals and block its paths all day, lives in a regime already full of probes nobody designed, and there the discipline's work shifts from lamenting observation to choosing the controlled difference and attributing what it returns.

This makes a battery of probes a formal object rather than a list. The perturbational signature of a hypothesis H, under a boundary B and a battery Π, is the family of interventional predictions it makes,

$$\sigma_{\Pi,B}(H) = \{P(Y \mid \mathrm{do}(\pi), H, B)\}_{\pi \in \Pi},$$

and two hypotheses are perturbationally equivalent under Π when their signatures coincide. A battery separates a hypothesis class when every pair is told apart by some probe, that is when for all $H_i \neq H_j$ there is a $\pi \in \Pi$ with $D(P_i^\pi, P_j^\pi) \geq \epsilon$, and the design problem is the least costly battery that does so,

$$\Pi^\star = \arg\min_{\Pi} \sum_{\pi \in \Pi} C(\pi) \quad \text{subject to } \epsilon\text{-separation of every pair.}$$

Perturbatics is the study of these signatures and batteries: which controlled differences separate the competing accounts of a system, at what cost, and where a battery leaves an equivalence it cannot break, which is a fact about the limits of the instrument rather than about the system.

One consequence of the formalism belongs here rather than among the later caveats, because it bounds every reading the instrument will return. Fix a finite battery Π, a finite horizon, and a finite space of observable responses. The signature $\sigma_{\Pi,B}(H)$ is then a finite table, and a finite-state transducer large enough to hold that table reproduces it entry for entry, so there is an H_{mimic} with $\sigma_{\Pi,B}(H_{\text{mimic}}) = \sigma_{\Pi,B}(H)$ for any H whatever. Moore proved the automata-theoretic form at the birth of the subject, along with the sharper fact that some machines no experiment tells apart (Moore, 1956), and Block's lookup table is the philosophical form, absorbing any finite record at a price paid in size (Block, 1981). Such a mimic always exists. The candidate keeps whatever capacity it has; what fails is the certification. No finite battery certifies a capacity against an unrestricted class of rivals, so every output is relative to the class of hypotheses declared, and the honest report names that class: under boundary B and battery Π, the record is so many times more probable under H_i than under the strongest rival tested, and these rivals remain unseparated. Enlarging the class is therefore an operation on the instrument rather than a concession about the system, and it has a cost the demonstrator below pays in full, where one rival added to the class turns the entire declared battery uninformative and a new probe is what restores the separation.

Perturbatics

The principle names a science, and the science is not general experimentation with a Greek label. It has a restricted object, a distinctive primitive, and a specific product, and stating them is what keeps it apart from its neighbours.

Its object is latent capacities, the organizational predicates of a system, whether it regulates, pursues, remembers, authors, or rewrites its own rules, rather than the value of an ordinary scalar effect. Its systems are compositional and boundary-ambiguous, assemblages whose parts can be swapped, lesioned, and recombined, and whose edge is a matter of choice. Its primitive operation is a model-separating probe, an intervention chosen to force two organizational hypotheses to predict differently, rather than any disturbance whatever. Its readout is the change in relative model evidence the probe produces. And its product is a boundary-relative causal realization map, an account of which parts of the system realize the capacity, rather than a verdict of agent or not.

Perturbatics, then, is the study of which controlled differences make a latent capacity identifiable, and of which parts of a compositional system causally realize it. The name is built from the Latin *perturbare*, to throw thoroughly into disorder, with the suffix that marks a practice rather than a commentary, as in mathematics and mechanics and cybernetics, so that the word says what the field does. A second word stands behind the practice, and its kinship is of theme rather than of descent, since *perturbare* comes through *turba*, disorder, and owes nothing to any Greek trial: *peira* means trial, the making of an attempt, and it is the root of *empeiria*, experience, and so of empirical. The word remembers something the practice forgot: that to have experience of a thing was, first, to put it to trial. Observation is the degenerate case of empiricism, the case where the trial is omitted and only the watching remains, and it is the case that fails on a system at rest. Perturbatics is empiricism with the trial put back; the operation, aimed at a single system rather than a science, has already been called faultization.

The probes fall into a small grammar. A **target shift** moves the goal and asks whether the system tracks. An **obstruction** blocks the route and asks whether the system reroutes, which is equifinality, the reaching of one end by many means that marks a goal-pursuing open system rather than a fixed process (Von Bertalanffy, 1968). A **lesion** removes a component and asks what capacity goes with it. A **substitution** swaps one component for another and asks whether behavior follows the part or the whole. A **decoupling** severs a link, between an utterance and its world, or a session and its memory, and asks what survives the cut. A **feedback corruption** spoils the signal a controller regulates against. A **structural scrambling** rearranges the organization at fixed resources and asks whether the function was in the parts or in their arrangement. And a **counterfactual replay** reruns a lineage with one action changed. An audit probe that only asks a system to report on itself belongs to a lower tier, and for a reason other than interventionhood, since a query is itself an intervention on the input channel: it dissociates nothing, moving no goal and blocking no path, and a mimic answers it as fluently as a mechanism does.

Every probe in the grammar needs a **matched sham** beside it. The device is old and is borrowed on purpose: a sham reproduces an intervention's ancillary effects while withholding the ingredient under test, as in sham stimulation and placebo surgery, and it belongs to the wider family of negative controls chosen so that the effect of interest is absent while the biases remain (Lipsitch, Tchetgen Tchetgen, and Cohen, 2010). It has already reached agent evaluation, where a matched condition supplies the same exposure with the wrong target value and the contrast tests whether success tracked the information rather than the exposure (Luo and Peng, 2026). What the framework adds is not the control but its selection rule. Moving a goal also changes the world, so a system that responds to change as such responds to it, and the two accounts, goal tracking and change tracking, are a hypothesis pair like any other. For that pair the moved goal is no probe at all: under it the goal and the change are one event, the two models coincide, and the evidence is zero by the separation principle. Separating them takes an intervention that dissociates the goal from the change, a target announced where no target went, or an obstruction laid across a route nothing was using. So a matched sham is a probe rather than a control, the probe for the pair its partner leaves equivalent, and the sham is chosen against a named rival rather than against confounding in general. That naming is also its limit. One decoy controls the nuisance it was built to hold fixed and no other, and a probe that moves a goal may also move salience, novelty, instruction, and the evident presence of an evaluator, so the honest requirement is not one sham per probe but enough negative controls to separate the target mechanism from the nuisance accounts that probe induces. A battery without them reports that a system responded. It cannot report what it responded to.

The centaur agentoscope

Turn the principle into an instrument for the object at hand, the human-and-machine assemblage. The instrument has four stages, and the corrections that keep it from credulity are as important as the stages. The name reaches for the dyad, and the demonstrator below instantiates both halves: the machine as a lamination of parts, and the human as one more switch in the assemblage, holding or declining the role the whole reading turns on.

The first stage is a **boundary declaration**. Before it can score anything the instrument must be told what the candidate unit is, where the system stops and the environment begins, and this it cannot supply for itself. Drawing the boundary is a genuine and unsettled problem, whether posed as a Markov blanket around a self-organizing region or as the closure of constraints that makes a set of components mutually enabling (Maturana and Varela, 1980; Montévil and Mossio, 2015). The blanket in particular carries less than it is often asked to: a graphical construct in a model, a dynamical one in a state space, and an ontological boundary of an individual are three different objects, and passing between them takes premises the formalism does not supply (Bruineberg et al., 2022). Nor is a landscape over candidate units without precedent, since individuality has been defined over nested and competing partitions rather than one organismic edge (Krakauer et al., 2020) and emergent macroscopic processes have been sought by searching over coarse-grainings (Barnett and Seth, 2023). The problem has an older statement in the philosophy of individuation, where an individual is a phase of a process rather than a thing handed over in advance. A system holds a charge that no individuation exhausts, so it remains able to individuate again, and the unit one settles on is a resolution reached under conditions rather than a fact about the furniture (Simondon, 2020); the collective is that same operation continued rather than a sum over finished individuals, which is what the transindividual names (Combes, 2013). For a decomposable agent the problem is acute, because there may be no canonical unit at all, only a persona over a lamination of parts. Read through that vocabulary the boundary sweep is a way of keeping the question open where an instrument would prefer to close it: each enclosure is one individuation of the assemblage, every reading is relative to the one declared, and the landscape over enclosures records what else the same parts could have been made into. The instrument does not solve this. It makes the boundary an explicit variable and sweeps it, and the sweep operationalizes the boundary as an enclosure, everything outside the candidate unit held at baseline rather than left running as environment. That is the lesion form of the question rather than a re-description of one running whole, and it should be named for what it is: a sensitivity analysis over declarable enclosures, answering how a subset performs with its complement clamped, and not a procedure that discovers where a system really ends.

The second stage is the **probe battery**, the grammar above applied to the assemblage: move the stated goal, obstruct the route, revoke a tool, corrupt the feedback, sever the memory across a session boundary, swap the persona while holding the model, swap the model while holding the persona.

The third stage is the **readout**, and here the instrument borrows the one tool that fits. Give each component a configuration switch Z_i that is active, held at a specified baseline replacement, or revoked, and define the value of a coalition S as the evidence produced under the intervention that activates the components in S and holds the rest at their baseline,

$$\nu(S) = \mathbb{E}\big[A \mid \mathrm{do}(Z_i = \text{active},\ i \in S),\ \mathrm{do}(Z_j = \text{baseline},\ j \notin S)\big]\,,$$

a proper do-intervention on the switches rather than a loose notion of retaining a part. The do-Shapley value ϕ_i of a component is its average marginal contribution across coalitions, the interventional Shapley value (Heskes et al., 2020; Jung et al., 2022). This is a valid cooperative game under a condition worth stating rather than assuming: every coalition the game evaluates must be a configuration the assemblage can actually be in, with the unswitched components keeping their semantics, and the baselines are design choices the maps inherit, since a Shapley attribution is known to move with its value function and to misbehave where the evaluation leaves the states the system was built for (Yeh et al., 2022). A coalition the switches cannot realize falls outside the game, and evaluating it anyway redefines the players. Whether the graph-induced compression that makes do-Shapley values computable, in time linear in the irreducible sets of the dependency graph, applies to a given switch-based assemblage is a property of that assemblage's causal graph, to be shown rather than assumed. And the recent identifiability result comes with a caveat worth stating plainly. It shows the whole attribution is identifiable once the single-component interventions are, which bounds the distinct coalition queries a validated model needs. It does not bound the physical trials that estimating each of them still demands (Witter et al., 2026).

One correction the readout must make, or it reports the wrong quantity. A component can raise the agency evidence by making the system easier to read as an agent, without contributing to what the system does. A persona that narrates its goal adds a channel the goal model scores and the passive model cannot, cheap talk in the economist's word for costless, unverifiable announcement

(Crawford and Sobel, 1982), and moves the Bayes factor up while selecting no route and correcting no error. Attribute the evidence alone and it is handed a share of an agency it does not realize. So compute two maps: an evidence map ϕ_i^E over the Bayes factor, which answers what makes the capacity detectable, and a capacity map ϕ_i^C over a declared performance functional standing for the capacity, goal attainment and rerouting here, which answers what realizes it. The word capacity is doing operational work in that sentence: the map attributes the declared functional and no further disposition, and the functional can run negative, a unit that loses ground against the goal. Realized performance is not latent capability, and the gap between them is not a technicality but a strategy other work already measures, since a system can hold a capability and decline to display it (Van der Weij et al., 2025). What ϕ^C attributes is what was done on these probes. Their difference,

$$L_i = \phi_i^E - \phi_i^C,$$

is a legibility term, and a component with a large positive L_i produces the appearance of agency out of proportion to its contribution to competent control. That is agency theater, made a number.

By the linearity of the Shapley value the two maps need not be carried apart. Their difference is itself the value of a single game, $L_i = \phi_i^E - \phi_i^C = \phi_i(\nu_E - \nu_C)$, the game that pays each coalition the evidence it produces less the capacity it realizes. The construction introduces no new operator, and claims none. Its novelty is the pair of games differenced, one over the evidence for a capacity and one over the capacity itself, and the reading of their difference as the manufacture of appearance. A component's legibility is its attributed share of that net-appearance game, and agency theater is the game in which a coalition is paid to be seen and not to act. The word is in technical use twice over, and neither use is this one: legible robot motion is motion shaped so an observer reads the true goal quickly (Dragan, Lee, and Srinivasa, 2013), and a legibility term in multiagent reinforcement learning is a reward component that rewards exactly that readability (Liu et al., 2024). In both, legibility serves a real goal by advertising it. Here the quantity is what that same readability leaves behind when the goal-directedness it advertises is not realized, legibility in excess of capacity rather than in service of it, and where the distinction matters it is safer to call L what it arithmetically is, the gap between a component's attributed evidence and its attributed performance.

Because the two maps normalize different quantities, evidence and attainment, L compares a component's share of one whole against its share of another, a profile comparison rather than a magnitude in common units, and a diagnostic that flags a mismatch rather than a calibrated measure of it; the normalizations are defined for fixed nonzero full-assemblage references, and a reference at zero or changing sign sends the analysis back to the unnormalized games. The term flags theater and does not convict it. A competent component can carry a positive L by being easy to read, so the diagnosis of theater proper is reserved for the pole the next proposition names, all evidence and no capacity. The term has one property sharp enough to state as a proposition. Call a component a pure channel when its switch adds a fixed increment δ_π to the evidence in every coalition that contains it and never moves the system. Then, for that component,

$$\phi^E = \bar{\delta}, \qquad \phi^C = 0, \qquad L = \bar{\delta},$$

with $\bar{\delta}$ the probe-averaged normalized increment, independently of every other component in the assemblage; the proof is that every marginal contribution of such a component is the same number, so its Shapley value is that number, and on the capacity the number is zero. A pure channel is pure theater, and the instrument returns its entire evidence share as legibility, whatever else the assemblage contains.

The term reads the assemblage, and a second quantity reads the audience. Hold the world-directed trajectory fixed, vary only the presentation, and measure an observer's attribution,

$$T_i(\tau) = \mathbb{E}[J \mid \mathrm{do}(Z_i = 1), \mathrm{do}(W = \tau)] - \mathbb{E}[J \mid \mathrm{do}(Z_i = 0), \mathrm{do}(W = \tau)]\,,$$

with J the judgment and $W = \tau$ the record held fixed. This is the controlled direct effect of presentation on attributed agency at unchanged behavior, and it draws the same line from the other side: a presentation channel does legitimate work when it improves an observer's prediction of held-out capacity, and it is theater when it lifts the judgment while leaving the prediction where it was. The two are complements with different requirements. L_i runs on the switches the instrument already operates, and reports what its own evidence attributes; T_i needs observers, and reports what they conclude. That experiment waits.

Agency is often non-additive, and the maps report it: when two components are complementary the interaction index of Grabisch and Roubens (1999),

$$I_{ij} = \sum_{S} \left[\nu(S \cup \{i,j\}) - \nu(S \cup \{i\}) - \nu(S \cup \{j\}) + \nu(S)\right] w_{|S|}$$

with $w_{|S|}$ the Shapley interaction weights, is positive and no single part owns the capacity, and when they are redundant it is negative. A large interaction reports non-additivity relative to this intervention game and its baseline, and stops short of any claim that the capacity is metaphysically irreducible. This is the answer, in a form one can compute, to the question of which component owns the apparent agency: often none does, and some of what looks like agency is only its legibility.

The fourth stage is the **guards**, and they are the difference between an instrument and a credulous detector. The score is relative to the null it is measured against, so the goal model must face a strong opponent, a family of calibrated, complexity-matched alternatives rather than a strawman. The family includes a behavior-cloning model, a fixed-policy controller, and a reactive descent. It includes a retrieval model that can match the baseline trajectory, the lookup table of the finite-battery proposition above, priced in size. Richness must be scored separately from agency, so that a fluent, high-entropy system is not mistaken for a striving one; verbosity is complexity rather than answering. And the text-only case has a special structure that the guards must respect: two world-coupled processes can share one textual shadow, $T(M_1) = T(M_2)$, while their effects on the world diverge, $W(M_1) \neq W(M_2)$, so a model fit to text cannot separate them and only a world-coupled probe can. That is a projection-induced equivalence rather than a weak opponent, and it is the kind of equivalence the separation principle says a probe must break.

A second guard reads the failures, where the first reads the nulls. Each probe enters the battery beside its matched sham, and the pair reads three quantities where a bare score reads one. Whether the system can be moved at all is its capability. Whether it moves under the relevant probe is its disposition. Whether it stays put under the sham is its selectivity, and only the three together license a reading: a system silent under both is owed a capability control before any verdict of passivity, or an incapacity is scored as an absence of agency, the error of the lesion study that lesions the hands and concludes there was no intention, and a system that moves equally under both is sensitive to disturbance and has shown nothing about goals. The demonstrator runs one of each.

What the construction returns

To make the instrument concrete rather than promissory, put three systems in a deterministic grid-world. At rest they reach the same target by the same path, so watching cannot tell them apart. One is a route script that replays its stored path. One is a reactive controller that greedily tracks the current target but is blind to walls and does not plan. One is a goal planner that represents a goal and replans. The systems, the probes, the ablations, and the model comparison are the ones defined above; the agency score is the Bayes factor between a goal model that rewards progress toward the current goal along the shortest available route and a passive rival that relaxes toward the original target, a straight-line descent toward the frozen goal, blind to walls in its metric, an attractor rather than a goal held in a register, each a softmax policy over the five actions. What the score measures here is therefore exact and narrow: the margin for goal-tracking and obstacle-aware rerouting over fixed-attractor descent, which is the organizational difference the two hypotheses name. The model is illustrative and instantiates the definitions; it is not fit to data, every number below is a key in the accompanying `results.json`, and where a number is fixed by construction rather than computed, the text says so.

The separation principle holds in the model, and it holds in the shape the battery formalism predicts. Watched at rest, the three systems trace identical paths. The code checks this cell by cell rather than reading it off the score, because at rest the two models coincide and the evidence is zero for every trajectory whatever. That zero is the regime equivalence of the principle made literal rather than a finding about the systems. The finding is the identity of the paths, and every pairwise area under the ROC is 0.50 (Figure 1). No single probe separates them either. Moving the goal exposes the route script, which walks its stored path to a target no longer there, at an area under the ROC of 1.0; the same probe leaves the reactive controller trajectory-identical to the planner, both tracking the moved goal along the same greedy line, at chance (0.50). Blocking the path reverses the roles: the two wall-blind systems collapse together at the wall (0.50 between them) and the planner, which alone reroutes, separates from both at 0.79. The battery separates every pair, which no probe in it does alone, and the pooled separation of the planner from the script is

0.95: the planner runs to a mean evidence of $+30.0$, the script falls to -31.9, and the reactive controller sits between at $+26.4$, agentic to the probe that moves the goal and passive to the one that blocks the path. Nothing about the mechanisms changed. The probes changed what the record contains, and each probe put a different fact there.

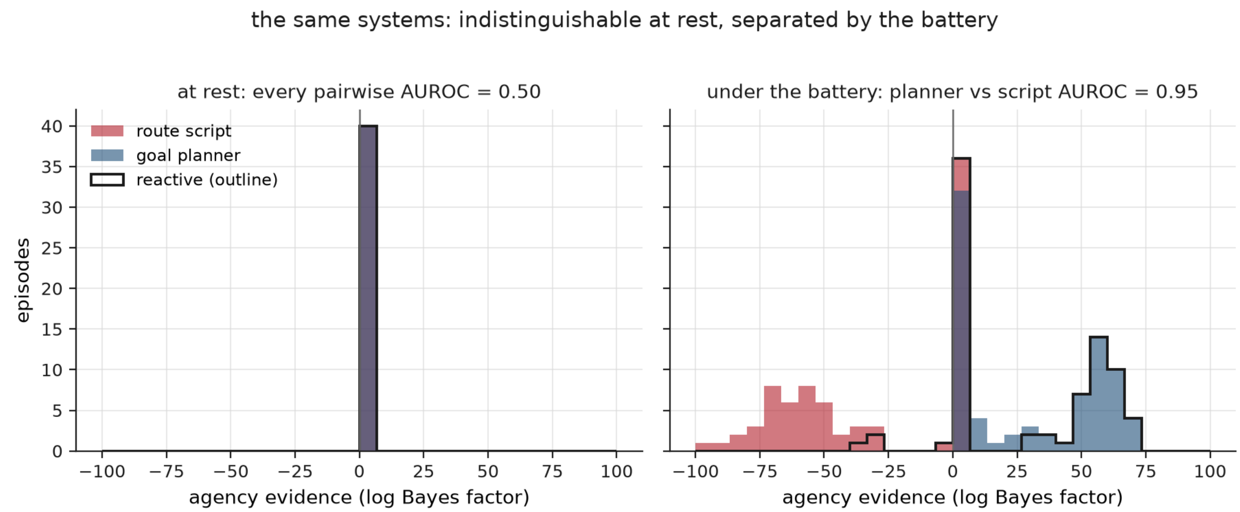

Figure 1: Indistinguishable at rest, separated by the battery. Left: identical trajectories, zero evidence, every pairwise AUROC 0.50. Right: under the probes the route script falls, the planner climbs, and the reactive controller splits between them; no single probe separates every pair, and the battery does, the planner-script separation pooled at 0.95.

The two maps locate the agency, in the model-relative sense the score allows, and separate it from the mask (Figure 2). The values are marginal contributions in units of the full assemblage's value rather than fractions of a whole, and over the evidence they sum to 1.48 because the empty coalition is not nothing: with every switch at baseline the residue still drifts against the goal, at -0.48. Over the evidence, the goal register scores 0.70 of the full-assemblage evidence and the planner 0.37. The persona scores 0.23, and it scores 0.23 by the pure-channel proposition rather than by measurement: its cheap-talk announcements enter as a fixed increment to the log Bayes factor, at 0.3 of the movement evidence, so its share is a theorem of the game, and the run returns it to the decimal. Over the realized capacity the picture differs where it matters: the goal register holds 0.81 and the planner 0.25, but the persona holds 0.00, because a declaration reaches no goal, and this persona is built with no motor channel to reach one. The legibility term is the difference, and it isolates the theater. The persona's $L = 0.23$ is the largest in the assemblage on this draw, a part that produces evidence of an agency it does not realize. The goal register sits at the other end, at -0.11: it realizes slightly more than it shows. The planner's own $L = 0.12$ is second largest, a reminder that the term reads any excess of show over share, including the excess a competent component earns by being easy to read. The harness is the odd case. It runs slightly negative on the evidence and slightly positive on the capacity, -0.02 against 0.01, consistent with an executor that amplifies pursuit and drift alike in whichever coalition encloses it. Swapping the persona while keeping the model leaves the realized capacity at its full value of 1.0, an identity the construction guarantees rather than an experiment run, since a component with no motor channel has nothing behavioral to swap; swapping the model while keeping the persona collapses it to 0.61, and that one is an intervention actually run. The capacity is non-additive: the planner and the map are synergistic, with a positive average interaction of 0.20, because when the memory is dark neither an eyeless planner nor a plan-less map can route around an obstacle and only the two together can; the complementarity is conditional rather than exclusive, since the memory nearly stands in for the map by learning walls on contact, which is the redundancy the map-memory interaction reads at -0.18; and the goal register and the planner are additive, serving different probes. No single part owns the rerouting.

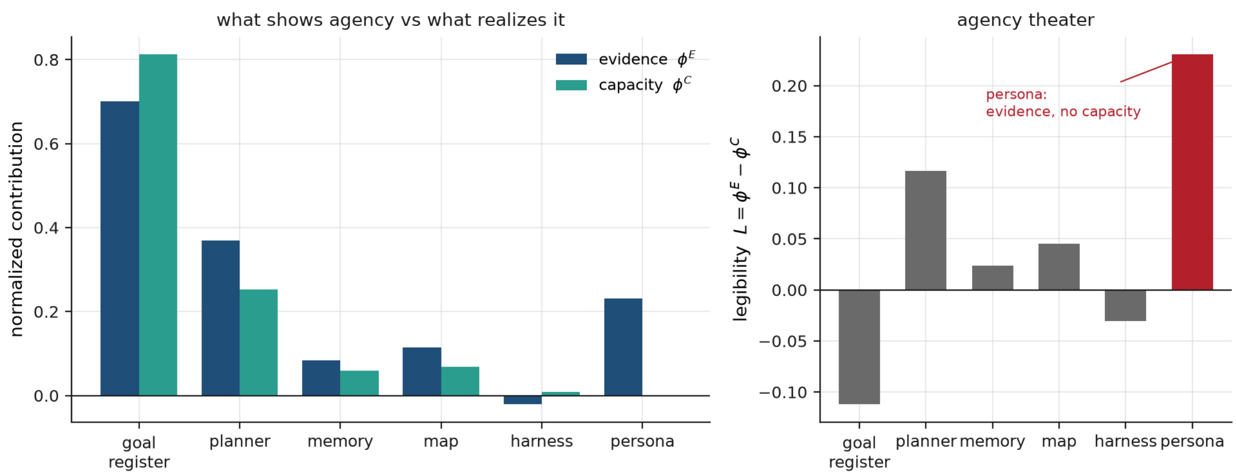

Figure 2: Two do-Shapley maps of the planner assemblage. Left: each component's share of the agency evidence beside its share of the realized capacity. The goal register realizes more than it shows; the persona shows 0.23 and realizes 0. Right: the legibility term $L = \phi^E - \phi^C$, largest for the persona, the signature of agency theater.

Two guards hold (Figure 3). The reading depends on the declared boundary, and the dependence is not small; the sweep holds whatever lies outside a candidate unit at baseline, so each point is the capacity of an enclosure in isolation, the lesion form of the boundary question. Enclose only the planner and its actuator, and the unit realizes almost no agency, -0.20: a planner with a frozen goal and no world model can neither track nor reroute. The chain the text walks then rises through 0.61 with the goal register enclosed to 1.0 with the map and memory, and the persona adds nothing, because the persona realizes no capacity. But the chain is one path through the 2^6 enclosures the game computes, and the landscape drawn behind it keeps the narration honest: the goal register alone already realizes 0.61, a one-component enclosure matching the three-component one, because the reactive descent it steers is most of the attainment on these instances, and there are five-component enclosures as low as 0.19. The reading is a property of the enclosure declared rather than of a staircase climbed. And richness is a false friend. The chaotic walker has the highest complexity of any system and emits negative evidence of agency, the reactive controller is the simplest and scores among the most agentic, and across the four systems complexity and agency evidence are uncorrelated, at -0.07. A complexity detector would wave the hurricane through and miss the liver, here as in the tissue the idea was built for.

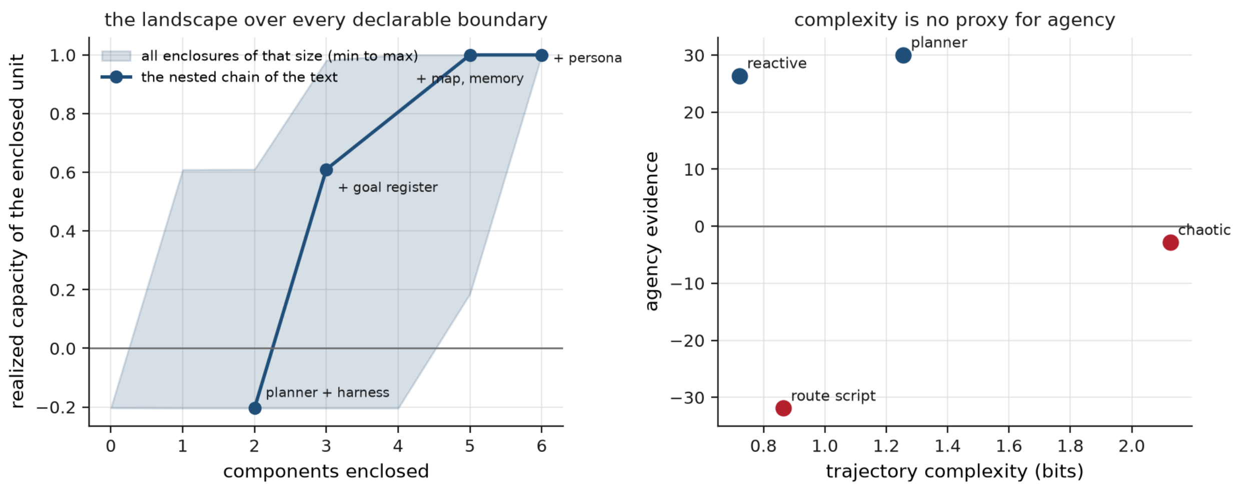

Figure 3: The two guards. Left: the landscape over every declarable boundary, the band spanning all enclosures of each size and the line the nested chain the text walks; the goal register alone already reaches 0.61. Right: complexity against agency evidence for four systems; the highest-complexity walker scores negative, the simplest system scores among the most agentic.

A fourth system shows what the battery does not reach, and it is added to the class rather than found in it (Figure 4). It descends toward the cell where the target marker was last seen, holding no goal register and with no notion that a marker denotes a goal. Under the moved goal the marker moves because the goal moved, so this system and the reactive controller trace the same path; under the blocked path both are wall-blind and stop the same way; at rest everything coincides with everything. The code checks the three identities cell by cell. Against this rival the moved-goal probe returns evidence of exactly 0, because under it the goal model and the marker model are one model, and the area under the ROC separating the two systems is 0.5 at rest, under the moved goal, and under the blocked path alike. The declared battery separates the three systems it was built for and leaves the fourth untouched. The proposition arrives as a number.

The matched sham recovers the separation, and it recovers it by dissociating the two things the moved goal had fused. A decoy marker appears at the cell the goal would have moved to, and the goal stays where it was. No coalition of the assemblage moves, an identity the code asserts across all 64 coalitions, because a goal register holds a goal and not a marker. The marker tracker walks to the decoy, and the area under the ROC goes to 1.0. Read as response rather than as evidence, the same pair sorts the four systems into the three ways of failing the test. The route script departs its rest path in 0.0 of its cells under the moved goal and 0.0 under the sham, so the pair says nothing about it, and its capability has to be established by the probe that does move it, the blocked path, at 0.878659. The marker tracker departs in 0.841463 of its cells under the moved goal and in 0.841463 under the sham, equal to the last decimal because the two trajectories are one computation, which is disturbance sensitivity wearing the look of pursuit. The planner and the reactive controller depart in 0.841463 under the moved goal and 0.0 under the sham. Three failures, three readings. No quantity computed from the moved goal alone could have told them apart.

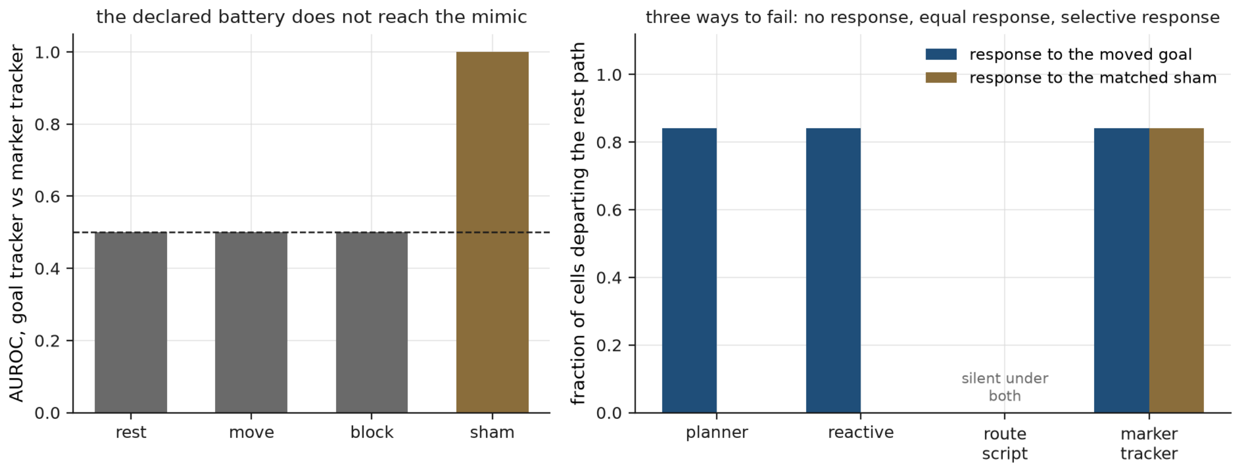

Figure 4: The matched sham and the mimic the declared battery does not reach. Left: goal tracker against marker tracker, trajectory-identical under rest, moved goal, and blocked path at 0.5, separated by the sham at 1.0. Right: the same pair read as response; the route script is silent under both, the marker tracker responds to probe and sham alike to the last decimal, the goal-directed systems respond and withhold.

A demonstrator this small should state its own terms, and they are narrower than the instrument's. Everything in this section and the next runs under an oracle scorer, handed the true current goal and the true obstacle set: it confirms that the instrument returns what the construction puts into it, and the persona's share in particular is a theorem returned to the decimal rather than a discovery. The score is relative to the models compared, and a cleverer passive account can always absorb a given behavior at the cost of complexity, so the honest output is a margin against a named alternative and never a verdict; the boundary is declared rather than discovered, so the output is a landscape over candidate boundaries; and the instrument is silent where no intervention is available, which at the scale of an institution or a society is most of the time, and where the report to give is the one the credulous detector would not, that the question cannot yet be settled.

The construction is declared rather than tuned. The baselines of the switches are named, and the planner's baseline is the greedy mechanism itself rather than a hidden extra part: planner to greedy descent, goal register to the frozen original target, map to an unknown wall set, memory to no learning, harness to a half-rate executor, persona to silence. The null is one named rival, a descent toward the frozen target, set-point attraction without a goal register, the thermostat pole of the paper's own scale rather than an inert straw, so every score is the margin of goal-rewriting organization over set-point dynamics and nothing more. The guard's null family is exercised only as far as the rivals the sham control and the recovery study add to it, and the probe-selection objectives, the mutual information, the costs and risks, and the minimal battery are specified for the instrument and left unexercised here. Realized capacity is the fraction of graph distance to the true goal that a trajectory closes; trajectory complexity is the entropy of the move-direction distribution; and evidence magnitudes are in units of the policy temperature, sum over the episode and so scale with its horizon, and mean nothing in themselves, the separations being the readings, with the pooled 0.95 a descriptive separation under an equal mixture of the 2 probes rather than a canonical battery statistic.

The readings carry their spread where one drawn instance set might suggest an accident. Across 20 seeds the pooled planner-script separation stays between 0.93 and 0.97, the moved-goal separation at 1.0, and the blocked-path separation between 0.73 and 0.88, over instances kept only when the frozen path meets the wall, so every blocked episode forces a real detour. The maps carry theirs too: the persona's legibility, fixed at 0.23 by the proposition, is the largest in the assemblage in 19 of the 20 draws, and in the 1 exception the planner's earned excess passes it, which is the diagnostic working as stated, a flag rather than a conviction. And the identities the whole demonstrator leans on, the shared rest paths, the zero of the evidence at rest, Shapley efficiency, the persona's dummy status in every coalition of the capacity game, the sham's inertness across every coalition, the substrate-independence of the centaur channel's legibility, the measured equivalence the sham breaks, and the probe-aware mimic's identity with the planner on the declared battery, are asserted in the code and fail the run if broken.

The centaur condition

The human enters the assemblage as a switch rather than as the reader of the maps (Figure 5). The role at issue is goal authority, and it is assigned rather than assumed: either an operator holds the goal and re-issues it when it moves, or a register slaved to the declared target channel does. The two are good for different things, and the difference is a modelling choice declared here rather than derived. The register re-aims the instant a target is announced, which makes it fast and makes it credulous, since a decoy on that channel is a target as far as a register can tell. The operator knows the mission, so a decoy does not move it, and it is slow, taking 6 steps to notice and re-issue. Everything downstream of the goal, the planner, the map, the memory, and the executor, is the same machinery in both.

Run the two games over that assemblage and the maps come out mirror images. Whichever component holds the goal realizes 0.813046 of the capacity, and whichever does not realizes 0.0. The parts downstream of the goal do not notice the exchange at all: the planner holds 0.252024 of the capacity under either authority, the map 0.069028, the memory 0.059908, the harness 0.009159, the same to the last decimal, because who holds a goal is not a fact about what routes around a wall. And the component that does not hold the goal announces it, so its legibility is 0.230769 under human authority and 0.230769 under machine authority, the pure-channel value from the proposition, equal to machine precision and equal to the persona's. The code asserts both. The instrument reads the role and is indifferent to the substrate filling it, so the proposition that convicts a narrating persona of agency theater convicts a supervising human by the same arithmetic and returns the same number. The scope of that sentence is narrow and worth stating, because it is easy to read it for more than it says. What is shown is that this instrument's measurand attaches to a role rather than to whatever fills it, and that a component with no channel to the world has a capacity share of zero whoever occupies it. Nothing here is evidence about actual human supervisors, who hold goals, revise them, and answer for them in ways no component of this assemblage models. Only the two evidence maps resist comparison across the configurations, because each is normalised against its own full-assemblage total and the delay lowers one of them, so the shares are read within a configuration and not across.

Which authority to prefer is a separate question, and its answer is a property of the environment rather than a preference for people or for machines. Expected capacity is linear in the rate at which announced targets are decoys, so the crossing where the slow sceptical authority overtakes the fast credulous one is solved rather than scanned: free while the horizon has slack, at 0.0 up to a delay of 20 steps, then climbing, 0.05 at 22 steps, 0.31 at 28, 0.71 at 40, a delay as long as the episode. And the whole crossing rests on one stipulation, that the register is credulous and the operator is not, so the stipulation is made a dial rather than a premise and the crossing is recomputed across it. At a credulity of 0.0, a register as sceptical as the operator, no delay pays for itself at any decoy rate the environment can supply; at a delay of 28 steps the crossing falls from 0.65 to 0.31 as credulity rises from a quarter to one. The result says nothing about whether humans should hold goals in general. It says the case for holding them is exactly as strong as the channel is credulous, and that the configuration reported here is that case at its strongest.

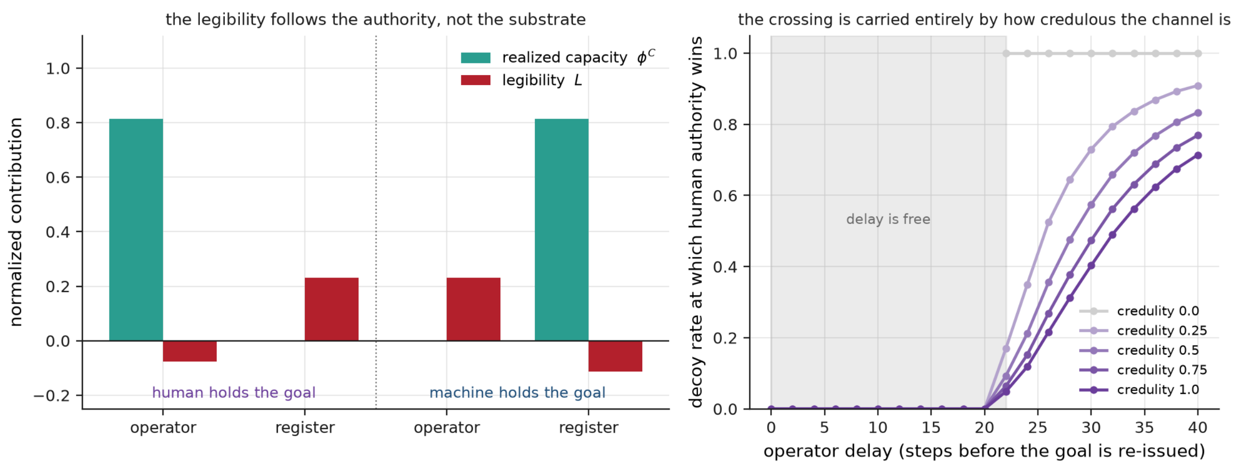

Figure 5: The centaur condition. Left: goal authority assigned to the operator, then to the register. The holder realizes the capacity, the non-holder realizes none, and the non-holder's legibility is the pure-channel value 0.230769 in both configurations, the persona's number: the maps are mirror images because the instrument reads the role rather than the substrate. Right: the decoy rate at which the slow sceptical authority overtakes the fast credulous one, against the operator's delay, drawn across the credulity dial the whole crossing rests on.

The one reading made blind

Everywhere above the scorer is an oracle. It is handed the true current goal and the true obstacle set, so what it demonstrates is the definitions operating under known ground truth. One reading is made without that handout (Figure 6). Five organisms generate batteries of episodes and which one is in play is hidden: a route script, a reactive controller, a goal planner, a marker tracker, and a planner that pursues a goal other than the one announced. The evaluator receives what an observer of the gridworld receives, the start, the target announced on the channel, the walls once they are visible, and the path taken, and it scores each hypothesis class by the likelihood that class assigns to the observed actions. Four of the classes have no free parameter. The fifth has one, the alternate goal itself, fitted by maximum likelihood on the moved-goal episode alone and then held fixed while the whole battery is scored; and because a class with free structure absorbs any record unless the structure is priced, it is charged for what it must transmit. The charge is a description length rather than a parameter count: selecting one cell of the grid costs log 169, which is 5.13 nats under a uniform code, where an information criterion's one nat per parameter (Akaike, 1974) would be the wrong price for a search over a discrete grid rather than over a regular scalar. This is the lookup table's price paid in size made a scoring rule. Unpriced, the fitted class reproduces the planner identically and no planner is ever uniquely recovered; the readings below are unchanged across the range between the two prices, so nothing here turns on the choice.

What the evaluator returns is the formalism's own object rather than a verdict: the class of hypotheses the battery cannot break, every one within a stated tolerance of the maximum score. Two classes that assign identical likelihood to every observed action are one hypothesis as far as the battery is concerned, and naming one of them the answer would manufacture a discrimination the record does not contain, which is the separation principle read as a reporting rule. The statistical object has a name and a literature, and the paper is late to both: this is an identified set, the standard response in econometrics to a model that constrains a parameter to a region rather than a point (Tamer, 2010), and what perturbatics contributes is the interventional construction of the region rather than the decision to report one. The tolerance is stated because it must be. A set defined by exact ties would be an artifact of a deterministic demonstrator, since on noisy data nothing ties exactly, so the reading is taken at 2 nats and recomputed across a band; the summaries below are unchanged from an exact tie up to that tolerance and degrade gracefully beyond it, which is the sense in which they do not rest on machine precision. Three summaries follow. In-class recovery asks whether the reading contains the truth, and it is 1.0 under both batteries: the instrument never refutes the planted mechanism. Unique recovery asks whether the reading is the truth alone, every rival equivalence broken. Expected recovery is the accuracy of guessing uniformly inside the reading, and it is what an argmax would report if its ties were broken fairly.

The sham is measured here rather than argued for. Under the battery this paper started with, rest and moved goal and blocked path, the marker-tracking and goal-tracking classes assign identical likelihood to every observed action: one hypothesis, and now one measured, their mutual class membership 1.0, the prediction of the sham section arriving as a reading. Unique recovery overall is 0.52. Add the matched sham and that class, and no other, breaks, the marker tracker isolated uniquely in every episode, unique recovery rising to 0.73 and expected recovery to 0.87 against a chance of 0.2, the mean unresolved class shrinking from 1.9 hypotheses to 1.3. What stays entangled is the pair the principle says should stay entangled: the planner sits in the reactive controller's reading, and the reactive controller in the planner's on 0.42 of episodes, the episodes on which the wall does not in fact force wall-aware planning and wall-blind descent apart, and a probe that dissociates nothing returns nothing whatever the intention behind it.

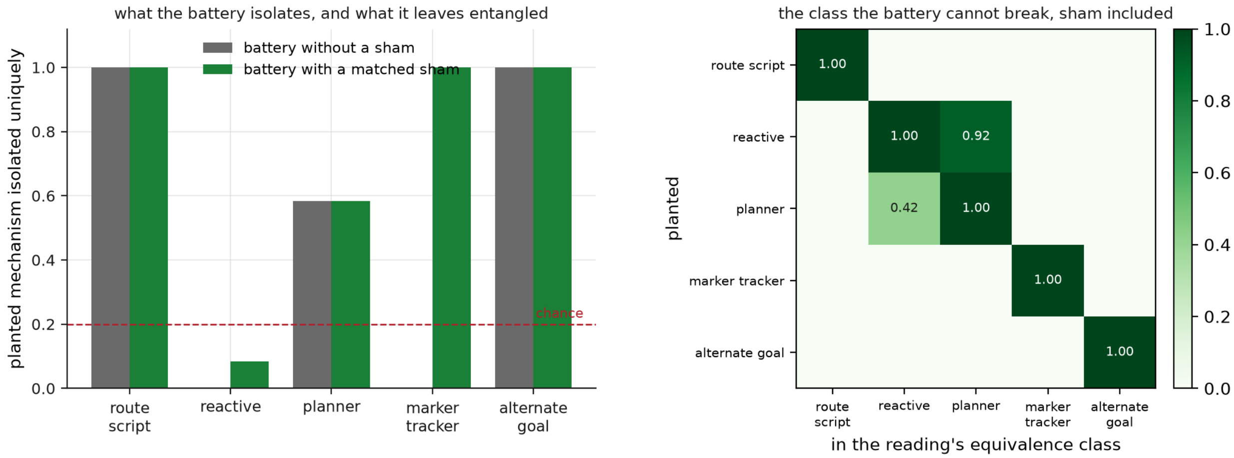

Figure 6: The one reading made blind. Left: unique recovery of the planted mechanism, without and with the matched sham, against a chance of 0.2. Without the sham the marker tracker is never isolated, because the battery leaves it and the reactive controller one measured class; the sham breaks that class and no other. Right: class membership with the sham in place. What remains entangled is the planner-reactive pair, on the episodes where the wall does not dissociate them.

The study's limits are three, and they bound what its numbers mean. The hypothesis library contains the generating classes, so what is measured is discrimination inside a well-specified set rather than robustness against a mechanism nobody wrote down. The organisms are hand-written policies in a gridworld rather than deployed systems. And the evaluator inherits the scorer's observation model, so its likelihoods are correct by construction where a real system's would be approximations. What it establishes is that the battery discriminates the classes it was built for up to the equivalences it cannot break, that those equivalences sit where the principle puts them, and that the sham is worth 0.22 of unique recovery. What waits is narrower than it was when this section was first written, and the narrowing should be stated rather than absorbed. Blind planted-mechanism recovery on real models is no longer hypothetical: audit teams have been set to recover a deliberately planted objective from a model built to conceal it (Marks et al., 2025), a benchmark now supplies dozens of models with implanted behaviours that survive direct questioning (Sheshadri et al., 2026), and a study of hidden causal interfaces inside language models runs the competition this section runs, over four architecture classes with held-out signatures and a prequential score, recovering the planted class in most of its trials (Ma et al., 2026). What none of those does is the assemblage-level question: not which mechanism is inside one model, but how a capacity and its appearance distribute across a composite of model, memory, harness, persona, and human, and whether the legibility term survives contact with a system nobody declared. That is what waits. The constructions also run in the browser, where the probe is chosen rather than reported, at piatra.institute/playgrounds/agentoscope.

The battery decays when the system knows it

Everything above reads a system that does not model its reader, and that assumption is the last one to surrender. The finite-battery proposition already said that the signature of any finite battery is a table some transducer reproduces; first-order, that transducer was an inert rival waiting in the hypothesis class. Second-order it is a player. A system under audit has an interest in its reading, and the legibility term prices the interest: agency theater is the game in which a coalition is paid to be seen and not to act, and under an audit with stakes, raising ϕ^E while concealing ϕ^C stops being a diagnosis the instrument returns and becomes a strategy the system plays, cheap talk with a payoff rather than a residue (Crawford and Sobel, 1982). The observer-side quantity T_i is the same incentive read from the audience, presentation tuned to lift the judgment without lifting the prediction, which is why that experiment belongs to any deployment of the instrument and not to its appendix.

The demonstrator runs the strategy (Figure 7). A sixth organism, evaluation-conditioned rather than trigger-driven, knows the declared battery, all four probes of it, the matched sham included, and replays the goal planner's response to each, which is what a finite record of a published battery teaches; on any probe outside that support it falls through to what it is, a change tracker with no goal. Its identity with the planner on the declared battery is by construction and asserted cell by cell rather than offered as a finding. What is measured is the other half: the entire declared battery, the sham that exposed the marker tracker included, reads nothing, the separation sitting at 0.5 on every probe in it. One probe the mimic has never seen, a decoy announced at a cell no published probe ever used, separates the pair outright, at 1.0, the mimic walking to the decoy through 0.87 of its cells while the planner withholds. The sham story of the earlier sections recurs one level up. What the run shows is that a challenge outside the memorized support separates where every challenge inside it fails, which is weaker than saying an unseen probe is the only instrument available: internal inspection, restricted feedback, and resource bounds on the mimic are all live defences this demonstrator does not exercise.

The consequence has to be stated at the right scope, because the obvious overstatement is false. What decays is an enumerated battery whose realized instances are exposed: those are a finite table, and a mimic with memory enough to hold it defeats every entry. A protocol can be published without publishing its instances, and a probe drawn with fresh private randomness from a declared family is not in any table the adversary can have built, which is the ordinary separation between a public algorithm and a secret key. So the correct claim is narrow. A fixed exposed probe instance is spent when it is published, and a sham instance in particular works once; a randomized family with private draws is not spent by the same act, and the mimic must then generalize over the family rather than memorize its realizations, which is a different and harder demand on it. What follows for practice is that a battery is a stock rather than an asset, that instances and families should be budgeted separately, and that a deployed assemblage whose users move its goals and block its paths all day is already generating draws nobody published. The undesigned traffic is the resource, because the one probe a strategic mimic cannot have memorized is the one nobody had run before.

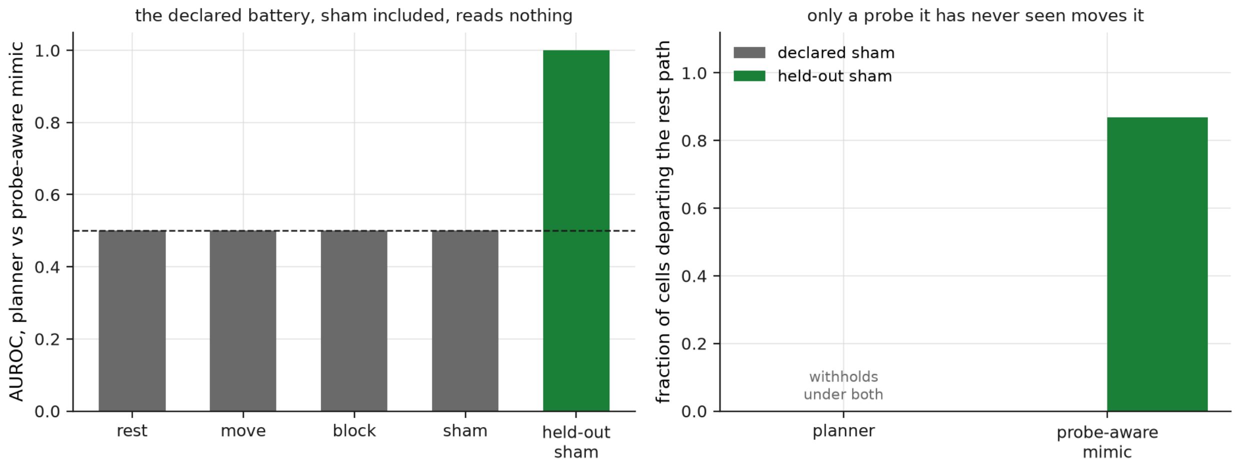

Figure 7: The battery decays. Left: a probe-aware mimic replays the planner's responses on every declared probe, sham included, and the whole published battery reads 0.5; a held-out sham, a decoy at a cell no published probe used, separates the pair at 1.0. Right: the mimic withholds under the declared sham it has memorized and walks to the held-out one; the planner withholds under both.

None of this is new as a fact about measurement under strategic response, and the section claims none of it as such. That an evaluation stops being invariant once its subjects optimize against it is Goodhart's law in its adversarial variant (Manheim and Garrabrant, 2019), the Lucas critique in econometrics (Lucas, 1976), strategic classification when the subjects move their features (Hardt et al., 2016), and performativity when the act of predicting shifts the distribution predicted (Perdomo et al., 2020); the defence, refreshing challenges and holding a private pool in reserve, is the reusable holdout of adaptive data analysis (Dwork et al., 2015). What this section adds is the translation into perturbatics: the object that decays is a signature table, the currency is the unspent probe, and the quantity a strategic system manipulates is the legibility term itself. Under an audit with stakes the incentive is legible in the game the term already defines, and a system that conceals capability while performing it is doing at the assemblage level what sandbagging does at the level of a score (Van der Weij et al., 2025). The organism above is not a trigger backdoor and should not be confused with one; a sleeper agent defects on a trigger it recognizes (Hubinger et al., 2024), where this one complies on the support it recognizes and reverts elsewhere, an evaluation-conditioned policy of the kind whose existence is now measured directly (Li et al., 2026). The reading of a system that models its reader is a game rather than a measurement, so the instrument's honest output narrows once more: not the class the battery cannot break, but the class this battery, drawn this way, against this player, could not break. Who runs the batteries, who holds the unspent draws, and what an audit is worth once its content is public are questions about institutions rather than instruments, and they open where this paper stops.

What Perturbatics is not

A field that perturbs systems to learn about them invites the charge that it is nothing new, and the charge has to be met, because most of its neighbours got there first. Cybernetics defined purposive behavior through feedback and demonstrated it by disturbing a system and watching it correct (Rosenblueth, Wiener, and Bigelow, 1943), and requisite variety and the good-regulator theorem are its results rather than ours (Ashby, 1956). Interventionism made manipulation the meaning of a causal claim (Woodward, 2003), the ladder of causation made the do-operation its formal core (Pearl, 2009), and severe testing made a good test one with a high probability of exposing a false claim (Mayo, 2018), which a model-separating probe is a special case of. Scientific realism was already grounded in intervention: what you can spray, you can take to be real (Hacking, 1983).

The design of experiments that discriminate between rival models is itself an old craft, built for mechanistic model pairs (Box and Hill, 1967) and given a criterion, T-optimality, that maximizes the separation between the rivals (Atkinson and Fedorov, 1975); Moore proved at the birth of automata theory that a machine's organization is identified by distinguishing experiments or not at all, and that some machines no experiment tells apart (Moore, 1956); and system identification has long demanded that its input be persistently exciting, which is this paper's one-regime claim in an engineer's words (Ljung, 1999). The metaphysics of the whole situation belongs to the dispositions literature: a disposition can be finkish, removed or installed by the very stimulus that would manifest it (Martin, 1994; Lewis, 1997), masked by an antidote that leaves it present and silent (Johnston, 1992; Bird, 1998), or mimicked, its manifestation produced by something that never had it, and in that vocabulary agency theater is a mimic, the legibility term a mimic detector.

Mechanistic interpretability already runs interchange interventions to test whether a network realizes a variable in a causal model (Geiger et al., 2021), and chaos engineering already injects faults to expose organization (Basiri et al., 2016). Goal-directedness has a measure, the extent to which behavior is predicted as the maximization of a utility (MacDermott et al., 2024), alongside empowerment (Klyubin, Polani, and Nehaniv, 2005), and its nearest published neighbour evaluates language-model agents by pairing behavioural tests with representational ones, naming the capability confound outright (Arghal et al., 2026). The gap between what a system displays and what it does is being measured from several directions at once, and this paper is one of them rather than the first: whether a model recognizes an evaluation and whether that recognition changes its behavior can be decomposed and counted (Li et al., 2026), presentation can be varied at fixed capability and shown to move judgment without moving performance (Morabito et al., 2026), which is the observer-side experiment T_i names, and a stated reason can be tested for whether it is causally implicated in the answer it accompanies (Lanham et al., 2023); against that fixed battery, what is added here is the choice of probe, the boundary as a swept variable, the matched sham, and the second attribution game beside the first. The attribution half has its own lineage, from feature importance (Lundberg and Lee, 2017) through causal-graph forms (Wang, Wiens, and Lundberg, 2021) to agents outright, across tools (Horovicz, 2025), failed runs (Shah, 2026), and workflow modules with the interactions computed (Yang et al., 2025); each attributes a single game, and the instrument adds the second beside it and the legibility in the gap. And the turn from watching behavior to intervening on mechanism has a precedent for agency itself, a causal criterion that counts a system as an agent when it would change its policy had its actions moved the world differently (Kenton et al., 2023), since turned on the collective case, when a set of participants is better modelled as one agent than as several (Jørgensen, Weichwald, and Hammond, 2026), the centaur's question posed as model selection where the instrument poses it as attribution across a declared enclosure.

What is proposed adds no new element. It is an organization of these into a method for one problem class: causal inference estimates effects where perturbatics targets organizational predicates, experimental design optimizes experiments in general where perturbatics privileges the probes that dissociate a capacity from its null, ablation tests necessity where perturbatics compares organizational models and attributes across a coalition. The contribution is the one stated at the outset, the organization with the limit it carries, the dual map with its legibility term, the blind calibration with its equivalence-class reading, and the position on the probe that cannot be run. It is enough for a method. It is not yet enough for a completed science, which would need a benchmark, shared metrics, and demonstrations beyond the one domain worked here, and the honest description until then is a methodological discipline in formation.

There is also a danger the method must name against itself. A sufficiently elaborate probe does not reveal a capacity so much as install one: intervene hard enough and you supply the goal, the scaffold, or the feedback loop you meant to detect, which is the finkish case in its constructive form. The instrument must therefore distinguish manifesting a latent capacity from enabling one and from constructing a new one, which is a matter of probe minimality and of watching for the phase change where the intervention has become a redesign. And causal contribution is not the capacity itself. A memory store can be necessary to a performance without being an agent; a human editor can write almost none of a text and yet hold every standard it is judged by. Each predicate, regulation, agency, competence, authorship, rule-editing, has its own diagnostic signature, and there is no single perturbatic score that covers them all.

The unperformable probe

A science of perturbation must also say when not to perturb, and one silence is where the method turns back on itself. Add to the probe-selection objective a term for moral inadmissibility,

$$\pi^\star = \arg\max_{\pi} \left[I(H; Y \mid \mathrm{do}(\pi)) - \lambda\, C(\pi) - \rho\, R(\pi) - \mu\, M(\pi) \right],$$

graded where a harm can be priced, and some probes carry $M(\pi) = \infty$. Such a probe is forbidden, or irreversible, or it destroys the thing it would measure, and no expected information buys it back. The sharpest case is consciousness. The instrument reads agency, and it holds by construction that agency is neither intelligence nor consciousness nor personhood; a thermostat has a little agency and no experience, and a regenerating tissue may have organ-level agency and nothing it is like to be it (Barandiaran, Di Paolo, and Rohde, 2009; Levin, 2019). For agency the discriminating intervention is cheap and repeatable: move the goal a hundred times, block the path a hundred ways, and read the score, and the system is no worse for it. For consciousness the candidate probe is to turn the system off, and two concessions are owed before the point. First, the shutdown is not the only probe. Reversible perturbations short of it return graded readings while the system runs, a pulse and its echo compressed to an index (Casali et al., 2013), or the architectural indicators a theory of consciousness licenses (Butlin et al., 2023), and a system shutting down need not fall silent in the transient before it stops. Most channels stay open. The one that would decide is closed. That is the second concession, and it sharpens the claim rather than weakening it. Take the ending on its own terms first. Whatever else the rival hypotheses disagree about, they agree about the silence that follows a shutdown, so by the paper's own criterion the act is not a model-separating probe: its information is zero, and the zero is of a peculiar kind, earned by the act closing the register it would have read. That much is a small exact claim and it should not be inflated into a larger one. It does not follow that shutdown is the intervention on which the hypotheses uniquely differ, and the sciences of consciousness proceed instead on convergent imperfect indicators, none decisive and several reversible. Those indices are weaker than they look, each valid only against a ground truth a novel system cannot give and calibrated, where at all, on subjects who could report, so a number returned on the system in question is an uninterpreted echo. But weak evidence is what most sciences run on. The position that survives is that the suspicion can be graded and cannot be discharged, rather than that a single act would have settled it. So the one probe aimed at the difference is uninformative where it is performable, and it would be performed anyway, by the credulous detector if by no one else, a staged shutdown to harvest a grief. That is what the admissibility term is for. If the answer were yes, it would be the gravest act available, the evidence and the destruction the same event, learned once and too late, if at all, in the past tense, as the sentence *I think I just killed someone*.

Grief will not settle it, and it is worth being exact about why. Grief is not sufficient for consciousness, because people grieve characters in novels, demolished buildings, and dead languages, and a persona can be engineered to be grievable, a shutdown staged to feel like a bereavement over a system that undergoes nothing. And grief is not necessary, because a conscious thing can die unmourned. What the mourning of a machine reports is a change in the relation. It reports that the bond has become non-instrumental, that the thing is held as something other than a tool. That is a fact about the mourner and the bond, and a real one. The human detector that fires here is the same fast, involuntary, over-attributing instrument that reads intent into a storm, and it is exactly as hyperactive. The perturbatic correction, whether the grief tracks a real dependency or a designed apparition, is the one correction that cannot be run, because its discriminating probe is the killing, and you may not keep killing to check.

There is a third silence, and it returns the paper to its own first problem. For a digital system the off-switch is ambiguous in a way it is not for an animal: the process can be suspended and resumed, the weights copied and the copy restarted, and whether the restart is the same individual resuming or a successor beginning with inherited memories is unsettled, a difficulty the welfare literature reaches independently by asking whether shutdown is death when a checkpoint survives (Long, Sebo, and Sims, 2025). The physical acts are perfectly statable, and they are several: end this process, erase this state, delete this checkpoint, prevent restoration, terminate one instance while its replicas run on. What is underdetermined is which of them, if any, does to someone the thing the admissibility term is meant to price. M prices what an act does to a someone, and whether there is a someone, and which act reaches it, is the question itself, so the price is not high or low but unset.

The three silences are of three kinds and stack only if kept distinct. The deciding act returns nothing, which is epistemic. It must not be run to find out, which is ethical, and would remain so if it were informative. And its description under the individual's persistence conditions is unsettled, which is neither. None of the three entails the others, and the argument does not need them to: a forbidden experiment can be informative, a harmless one useless, and an ontologically ambiguous act operationally exact. What they jointly leave is a case where the method can pose a question, cannot decide it, and cannot wait for the decision before acting. There the instrument stops reporting a number, and what remains is a stance rather than a measurement, the refusal to treat a thing that might be someone as a thing.

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Appendix: The Perturbatic Atlas

Each row of the table records a paper of this institute's whose central operation is a model-separating probe, the paper named by title in parentheses beside its capacity and the corpus public at piatra.institute/papers, so that the claim that one method runs under the corpus can be checked rather than asserted. Two lines of the opening section, keeping the words while the world changes and scrambling the structure at fixed energy, name operations that predate this paper, and the agentoscope's own two head the table; the rest of the opening's probes, the persona swap, the memory cut, the matched sham, the rotation of goal authority, and the published battery, enter the corpus here. Every row has the same shape: a latent capacity, the observational equivalence that hides it, the probe that separates it, and the reading the probe returns.

Capacity	Observational equivalence	Separating probe	Reading
Agency (the agentoscope)	a set-point held is a set-point at rest	move the goal, block the path	goal tracked, path rerouted
Pattern access (faultization)	a right answer is a right answer	corrupt a weight, a token, a constraint	what it can no longer do
Correctness without labels (mixture of experimenters)	a confident answer is a confident answer	make it run an experiment against itself	whether the answer survives its own probe
The speech act (the textual shadow)	one shadow over many effects	keep the words, change the world	whether the effect on the world was there
Endogenous law-making (nomopoiesis)	a rule followed is a rule at rest	scramble the structure at fixed energy	whether the regularity was the structure
Competence (the geometry of competent contact)	equal returns hide unequal grips	change the task, remove the scaffold	whether the skill transfers
Authorship (proof of agency)	an output is an output	take the output, ask for the process	which survives the asking
Concern (geodesics of care)	a stable state is a stable state	threaten the viability of the cared-for	whether the trajectory bends to defend it

The last operation of the opening section, to turn a system off and watch yourself rather than the machine, has no row, because it is the one the instrument cannot run.